

significant changes in characteristics.

The high-temperature end presents more difficulty. Practically, the upper temperature limit is determined by many factors and for semiconductor devices, it often does not reflect the inherent temperature limit of the semiconductor material. The limit is frequently determined by the interconnections and packaging, both for active devices and passive components.

As an indication of the practical upper limit, circuits have been offered commercially for operation up to +300°C.

There is also interest in applying HTE to aircraft, both commercial and military, particularly with the trend towards 'fly-by-wire' and distributed systems. Electronic interface circuitry would be co-located with monitoring and control transducers for engines, actuators and other aircraft system elements. Many of these locations involve a hot environment, possibly up to +300°C.

A similar situation arises for low temperatures. A thriving application is for scientific and military spacecrafts. Many sensors used for astronomical observations or surveillance, such as infrared or X-ray detectors, must operate at very low temperatures. In order to extract the ultimate performance from these sensors, it is necessary to co-locate the initial signal-processing electronics with the sensors in the cold environment. Generally, this environment is in the cryogenic temperature range.

Moreover, since very little of the solar system falls within the conventional-electronics temperature range, ETE might be a key technology for exploration of the solar system. High-temperature electronics will increasingly penetrate the market for combustion monitoring and control; particularly where distributed systems and/or improved performance are required. It will enable sensors with increased functionality to be placed closer to the actual point of combustion, which, in turn, will allow improved combustion efficiency.

Developments

Use of diamond for HTE. Because of their exceptional physical and chemical properties, diamonds are an invaluable semiconductor for manufacturing of electronic and optoelectronic devices. However, the fact that an n-type diamond with satisfactory electrical properties does not exist, is a serious handicap in the production of such devices.

Researchers have recently demonstrated for the first time that it is possible to obtain an n-type diamond with high electrical conductivity at room temperature. The operation consists of diffusing hydrogen into the boron-doped diamond. A reversible conversion from p-type into n-type is thus obtained. The type-'n' electrical

figures as high as 650°C. Silicon-based semiconductor electronics cannot function at these temperatures. The large performance gains made possible by high-temperature high-power capabilities of SiC offer economically large performance benefits to the aircraft, automotive, communications, power and spacecraft industries. The tremendous advantages of SiC electronics in these specific applications are slowly becoming a reality. Developing and maturing SiC technology to the point where it is ready for widespread system insertion is the focus of increasingly intense research efforts around the world.

Varying gravity method. Students from the UW-Madison, USA, designed and tested an experiment in varying

The usual motivation for operating electronics at extreme temperatures is improvement in the overall performance of a system by having part of the system—electronic sub-systems—operate outside the usual temperature range

conductivity values obtained are 10,000 times greater than those obtained using the standard process (direct incorporation of phosphorus).

Applications mainly concern high-temperature diamond-based electronics (transistors, diodes, etc) that will be necessary in the future in the fields of space exploration and the automotive industry, as well as telecommunications and energy distribution.

Use of carbide (SiC). Researchers are developing silicon carbide (SiC) as a material for advanced semiconductor electronic device applications. SiC-based electronics and sensors would operate in hostile environments (about 600°C) where conventional silicon-based electronics (limited to 350°C) cannot function. The ability of silicon carbide to function in high-temperature, high-power and high-radiation conditions will enable large performance enhancements to a wide variety of systems and applications.

SiC devices have repeatedly demonstrated proper operation at tempera-

ture conditions. The results from the experiment showed that a particular spray cooling method is not dependent on gravity—a significant finding that means spray cooling could be used in airplane and other high-heat electronics. The team developed a system that sprays dielectric liquid in a linear array directly onto the chips. The direct contact maximises the amount of heat transferred from the chip to the liquid.

Traditional methods direct the liquid upward, relying on gravity to drain the liquid away. However, this won't work in airplanes or spacecraft, which go through varying gravity conditions.

To be reliable, spray cooling should not be gravity-dependent. Linear spray array could be the answer, but testing was difficult. The team found that linear spray cooling is effective in both zero-gravity and double-gravity conditions. ●

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